



RESEARCH SYNTHESIS COMPLETE

I've synthesized extensive research across 8 major domains into a comprehensive guide that's ready for your product development. Here's what I've delivered:

Document Contents (16,000+ words):

Section A: Scientific Foundations

- ✓ Vocal prosody research: How pitch, pacing, pauses, and tone signal confidence & credibility
- ✓ Articulation/clarity science: How to measure and teach clear speech
- ✓ Emotion recognition in audio: Critical limitations and biases (gender, accent, age) you must address
- ✓ Evidence for communication training effectiveness

Section B: UX & Design Best Practices

- ✓ How to simplify complex metrics for consumers (avoid jargon, use comparisons)
- ✓ Visualization methods that work (time-series graphs, before/after, progress tracking)
- ✓ Dashboard structure: "How did I sound?" → "What should I do next?"
- ✓ Onboarding flows for unfamiliar metrics
- ✓ Gamification elements proven to increase retention (22% average lift)

Section C: Product Strategy

- ✓ 5 key user segments with jobs-to-be-done (job seekers, presenters, creators, non-native speakers)
- ✓ What consumers actually value (progress visibility, specific feedback, coaching not just metrics)
- ✓ Retention drivers vs. overwhelm triggers
- ✓ Pricing models: Freemium vs. course-based vs. B2B
- ✓ Your differentiation opportunities (volitional diagnostics, ethical AI, integrated coaching)

Section D: Technical Approaches

- ☒ Real-time audio processing (WebRTC, on-device vs. server-side trade-offs)
- ☒ Concrete algorithms for each metric:
 - Pacing/Speaking Rate (VAD + ASR-based counting)
 - Clarity/Articulation (formant extraction + spectral analysis)
 - Filler Words (ASR + pattern matching)
 - Energy/Confidence (pitch range + intonation + volume stability)
 - Pause Patterns (silence detection + classification)
- ☒ Recommended ML models (Whisper ASR, Silero VAD, CREPE pitch, etc.)
- ☒ Hybrid architecture recommendation (real-time on-device + server-side batch processing)

Section E: Competitor Analysis

- ☒ Detailed breakdown of Yoodli, Orai, Poised, Speeko (what they do well, user complaints, positioning)
- ☒ Market gaps (emotion detection transparency, non-native speaker focus, root-cause diagnosis, social confidence, aging population)
- ☒ UX mistakes to avoid (overwhelming metrics, judgment language, no onboarding, bias in emotion detection)

Section F: Behavior Change & Feedback Psychology

- ☒ GAIN Framework (Goal, Ally, Impact, Non-judgmental) – evidence-based feedback approach
- ☒ Growth mindset orientation (how to frame feedback for motivation, not shame)
- ☒ Psychological safety principles (Edmondson)
- ☒ Volitional diagnostics strategy (identify root cause: skill gap vs. motivation gap vs. commitment gap)
- ☒ How to prevent dropout and demotivation

Section G: Strategic Recommendations

- ☒ MVP feature roadmap (Phase 1: Core, Phase 2: Engagement, Phase 3: Differentiation)
 - ☒ 3 positioning options (empathy-driven, diagnostic-driven, inclusive-driven)
 - ☒ Go-to-market strategy (content marketing, partnerships, referral)
 - ☒ Monetization tiers (freemium + premium + optional coaching)
 - ☒ Success metrics to track (retention, conversion, NPS by phase)
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KEY STRATEGIC INSIGHTS FOR YOU

1. Your Unique Competitive Advantage

Your combination of:

- Clinical expertise (physician, multiple board certifications)
- Deep knowledge of volitional systems & behavior change psychology
- Ethical, transparent approach to AI
- Technical capability (Gemini Live API, no-code automation)

Differentiation Strategy:

- Don't compete on features with Orai/Yoodli (they're well-funded, feature-rich)
- Instead, build on your unique IP: Volitional diagnostics
 - "Why do people struggle?" (skill gap, motivation gap, commitment gap, etc.)
 - Personalized coaching based on root cause
 - Integrate optional human coaching (you or contractor network)
- Position as ethical: "The only speech coaching app that's transparent about emotion detection biases"

2. Emotion Recognition is a Trap

- Don't: Build emotion detection as a core feature (it's biased against women, non-native speakers, neurodivergent folks)
- Do: Focus on less-biased metrics first (pacing, clarity, fillers, energy as composite of non-emotional features)
- If you include emotion: Label it clearly as "experimental/beta" with disclaimers; frame as optional feature

3. Market Opportunity is Real but Requires Focus

- TAM is huge but fragmented: job seekers, presenters, creators, non-native speakers, dating/social confidence, older adults
- Don't try to serve everyone at launch
- Start with: Job seekers + interview prep (highest intent, high volume, willingness to pay)
- Then expand: Public speaking, creators, non-native speakers (in that order)

4. Freemium is the Right Model

- Free tier: Unlimited recordings + 3-4 basic metrics = builds habit, lowers friction
- Premium (\$10-15/mo): All metrics, transcripts, coaching videos, progress tracking
- Coaching add-on (\$50-100/mo): Optional human verification/check-ins (could be you initially)
- Target: 20%+ premium conversion rate (Orai/Yoodli achieve this)

5. Real-Time Feedback Wins

- Post-hoc analysis is easier technically but less engaging
- Real-time visual feedback (pacing gauge, filler word alerts) during recording drives engagement
- Poised's "live meeting feedback" is their killer feature
- Your app should include: Real-time pacing gauge + filler word alerts (doable with streaming ASR)

6. Gamification Works But Can Backfire

- Do use: Streaks (Duolingo effect: 12% engagement lift), progress charts, celebration of progress
- Avoid: Global leaderboards, badges for everything, judgment language
- Instead: Personalized benchmarks, comparison to own past self, effort-based rewards (not outcome-based)

7. Behavior Change Psychology is Core

- GAIN Framework beats judgment-based feedback (Lenny Pondy research)
- Transparency about limitations builds trust (especially with emotion detection)
- Volitional diagnostics is your edge: Help users identify the *real* blocker (not just "you're slow")
- Psychological safety matters: Users won't engage if feedback feels judgmental

IMMEDIATE ACTION ITEMS

Week 1: Validate Core Assumptions

- Interview 10-20 job seekers: What's their #1 pain point in interviews? (pacing? confidence? clarity?)
- Test assumption: Would they pay \$10-15/mo for coaching?
- Which metrics matter most in their experience?

Week 2-3: Build MVP Prototype

- Core metrics first (4 only): Pacing, Filler Words, Clarity, Energy
- Recording + transcript + dashboard (leverage your Gemini Live API experience for audio streaming)
- Free tier launch: 1 recording per day, basic metrics, no paywall for 30 days

Week 4-8: Validate + Iterate

- Get 500+ users through content marketing (blog posts on interview tips)
- Measure: 30-day retention (target: 30%), NPS (target: 40+), premium interest
- Iterate based on what users actually value

Month 3+: Scale Phase 1

- Add real-time feedback (pacing gauge, filler alerts)
 - Add scenario-based practice (interview Q&A scripts)
 - Micro-lessons (2-3 min coaching videos)
 - Launch premium tier
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DOCUMENT DOWNLOAD

Your comprehensive research document is saved as [speech-coaching-research.md](#) and includes:

- ✓ 60+ research citations (PMC journals, ISCA conference papers, industry reports)
 - ✓ Concrete algorithms for each metric (code-ready pseudocode)
 - ✓ Detailed competitor analysis (Yoodli, Orai, Poised, Speeko)
 - ✓ UX patterns from top apps + what to avoid
 - ✓ Technical architecture recommendations (hybrid real-time + batch)
 - ✓ Product roadmap (3-phase rollout)
 - ✓ Go-to-market strategy + positioning options
 - ✓ Monetization tiers + success metrics
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FINAL THOUGHT

You're in a unique position. Most speech coaching apps are built by product teams that understand metrics and gamification. You bring something rarer: deep knowledge of why people struggle to change, what actually motivates behavior improvement, and how to coach with psychological safety.

Your volitional systems framework could be a genuine differentiator. Instead of "optimize your pacing," you'd diagnose: "You have the skill but not the commitment—here's how to build habit."

That's not a feature. That's a philosophy. And it's what users actually need.